Step 5 : Calculation

Test statistic is computed as under:

Observed frequency (O_i)	Expected frequency (E_i)	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E_i}$
61	41	20	400	9.756
14	37	-23	529	14.297
10	16	-6	36	2.250
7	5			
5 } 15	1 } 6	9	81	13.5
3	0			
			Total	39.803

Note: In the above table, we find the cell frequencies 0,1 in the expected frequency column (E) is less than 5, Hence, we combine (pool) with either succeeding or preceding one such that the total is made greater than 5. Here we have pooled with preceding frequency 5 such that the total frequency is made greater than 5. Correspondingly, cell frequencies in observed frequencies are pooled.

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$= 39.803$$

Step 6 : Critical value

Degrees of freedom = $(k - 1 - s) = 4 - 1 - 1 = 2$

Critical value for 2 *d.f* at 5% level of significance is 5.991 *i.e.*, $\chi^2_{2,0.05} = 5.991$

Step 7 : Decision

The calculated $\chi_0^2 (=39.803)$ is greater than the critical value (5.991) at 5% level of significance. Hence, we reject H_0. i.e., fitting of Poisson distribution is not appropriate for the given data.

Example 2.13

A sample 800 students appeared for a competitive examination. It was found that 320 students have failed, 270 have secured a third grade, 190 have secured a second grade and the remaining students qualified in first grade. The general opinion that the above grades are in the ratio 4:3:2:1 respectively. Test the hypothesis the general opinion about the grades is appropriate at 5% level of significance.

Step 1 : Null hypothesis H_0: The result in four grades follows the ratio 4:3:2:1

Alternative hypothesis H_1: The result in four grades does not follows the ratio 4:3:2:1

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